

35. Consider the linear transformation $T: \mathbb{R}^5 \rightarrow \mathbb{R}^3$ given by

$$T(a, b, c, d, e) = (b + c + 4d + 2e, a + c + 3d + e, 4a - 3b + c - 2e).$$

- (a) Find a basis and determine the dimension of $\text{Ker}(T)$.
- (b) Find a basis and determine the dimension of $\text{Im}(T)$.
- (c) Determine the general form of $v \in \mathbb{R}^5$ such that $T(v) = (1, 2, 5)$.
- (d) Determine whether T is injective, surjective, bijective.
- (e) Determine whether T is invertible. If yes, give an explicit formula for T^{-1} .

For (d), give a solution that does not involve any further computations: the answer should follow directly from the answers to (a,b) by applying relevant propositions about these kinds of linear transformations.

36. Consider the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T(a, b, c) = (3b + c, 2a + b, a)$.

- (a) Determine whether T is invertible. If yes, give an explicit formula for T^{-1} .
- (b) Use the answer to (a) to find $\text{Ker}(T)$ and $\text{Im}(T)$. Give the simplest possible descriptions of these spaces.
- (c) Use the answer to (a) to determine whether T is injective, surjective, bijective.

37. Consider the linear transformation $T: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ given by $T(X) = \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix} X$.

- (a) Find a basis and determine the dimension of $\text{Ker}(T)$.
- (b) Find a basis and determine the dimension of $\text{Im}(T)$.

38. Determine whether there exists a linear transformation T that satisfies the given requirements. If not, prove that. If yes, give an explicit example and explain how you found it.

- (a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $\text{Ker}(T) = \text{Im}(T)$.
- (b) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $\text{Ker}(T) = \text{Im}(T)$.
- (c) $T: \mathbb{R}^6 \rightarrow \mathbb{R}^3$ such that $\dim(\text{Ker}(T)) = 2$.